



University
of Exeter

BEMM457: Topics in Business Analytics

Uncovering Patterns in Historical Stock Price Data: A Business Analytics Study on Reliance Equity

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GitHub Link: <https://github.com/aditimanwal/TIBA>

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Introduction

Background and Context:

Reliance Industries Limited (RIL) is one of India's largest and most prominent corporation, with operations spanning in multiple sectors including energy, petrochemicals, telecommunications, and retail. RIL plays a key role in the Indian stock market, with its equity shares actively traded on major exchanges. The company's stock performance not only reflects investor's confidence but also serves as a vital indicator of broader economic and market trends.

The dataset I have used in this study comprises of historical stock from 1st January,2023 to 29th November,2024 for Reliance Equity including daily metrics such as OPEN, CLOSE, HIGH, LOW, trading VOLUME, and VALUE. This data is publicly available and provides a rich foundation for analysis, enabling the exploration of stock price behaviour, trading activity, and underlying patterns.

Objective:

The primary objective of this report is to analyse the historical stock performance of reliance equity to uncover trends, relationships, and patterns in the data by using business analytics techniques. This report aims to draw actionable insights into the stock's behaviour over time and its relationship with trading metrics.

Relevance and Scope:

Stock price analysis is very crucial for investors, traders, and analysts who aims to make informed decisions. By understanding trends in price movements and trading volumes, stakeholders can identify opportunities, manage risks, and develop data-driven trading strategies. This analysis of Reliance Equity which is a widely tracked stock contributes to financial market research and provides valuable insights into price trends and market behaviour.

Research Questions:

- What are the trends in the daily stock price (OPEN, CLOSE, HIGH, and LOW) of Reliance Equity over the given period?
- How does trading volume influence stock price movements?
- What significant peaks or troughs can be identified, and what insights do they provide for trading strategies?
- What are the statistical relationships between price metrics and trading volume?
- What is the volatility in stock prices, and how can it inform investor risk management strategies?
- How does the stock price's moving average behave over time, and what trends can be observed?

Data Access, Ethics, and Preparation

Data Access and Source:

The dataset for this analysis is sourced from the National Stock Exchange of India's (NSE) official website, ensuring data authenticity, accuracy, and public accessibility. It contains daily historical stock data for Reliance Equity, one of India's most traded stocks. This dataset supports the analysis of stock price trends, volatility, and volume-price relationships. Its key variables include:

- Data: Trading date
- OPEN, HIGH, LOW, CLOSE: Daily stock prices
- VOLUME: Number of shares traded
- VALUE: Total monetary value of traded shares
- 52W H, 52W L: 52-week high and low prices

Data Cleaning and Pre-processing:

I started by importing necessary libraries for this project such as pandas, matplotlib, seaborn, and NumPy. Then I standardised the column names such as the Date column was formatted as datetime and removed whitespace from column headers for consistent reference, and missing/duplicate records were removed. Furthermore, I transformed the data by creating new metrics which includes Price Range (HIGH-LOW), Daily Change (CLOSE-OPEN), Moving Averages (7,30,50,100 day), Bollinger Bands. No scaling was required as financial data was analysed in raw form. Validated the data by descriptive statistical analysis, correlation analysis and outlier detection which supports in checking unusually large VOLUME or PRICE changes to ensure data integrity.

Ethical Considerations

The chosen dataset is publicly available and accessed via official source which holds none sensitive information. All preprocessing steps ensured transparency, and no manipulations were made to distort the data.

[3]:

	Date	series	OPEN	HIGH	LOW	PREV. CLOSE	ltp	close	vwap	52W H	52W L	VOLUME	VALUE	No of trades
0	2024-11-28	EQ	1289.95	1296.60	1269.05	1293.20	1272.75	1270.80	1280.74	3217.6	1217.25	15027412	1.924615e+10	311959
1	2024-11-27	EQ	1293.50	1303.85	1286.00	1295.70	1293.40	1293.20	1293.84	3217.6	1217.25	9572746	1.238561e+10	196614
2	2024-11-26	EQ	1294.95	1297.85	1284.20	1287.00	1295.60	1295.70	1292.03	3217.6	1217.25	9247557	1.194813e+10	218738
3	2024-11-25	EQ	1290.00	1304.45	1282.45	1265.40	1290.95	1287.00	1291.52	3217.6	1217.25	28923341	3.735509e+10	390544
4	2024-11-22	EQ	1229.40	1268.85	1222.45	1223.00	1264.00	1265.40	1247.09	3217.6	1217.25	20471957	2.553046e+10	323219
5	2024-11-21	EQ	1241.65	1243.95	1217.25	1241.65	1220.45	1223.00	1224.92	3217.6	1217.25	19433843	2.380490e+10	553410
6	2024-11-19	EQ	1260.75	1281.50	1235.15	1260.75	1237.40	1241.65	1265.50	3217.6	1235.15	14183746	1.794946e+10	355649
7	2024-11-18	EQ	1267.60	1272.90	1243.90	1267.60	1259.15	1260.75	1258.64	3217.6	1243.90	13620611	1.714350e+10	301906
8	2024-11-14	EQ	1253.60	1272.60	1251.10	1252.05	1266.90	1267.60	1263.73	3217.6	1249.50	12612633	1.593893e+10	301253
9	2024-11-13	EQ	1270.25	1275.45	1249.50	1274.25	1252.70	1252.05	1258.62	3217.6	1249.50	14633633	1.841821e+10	469215
10	2024-11-12	EQ	1275.55	1289.30	1267.50	1272.70	1269.50	1274.25	1277.82	3217.6	1267.00	10671091	1.363569e+10	299780

Figure 1: Loading and preparing the dataset

Exploratory Data Analysis (EDA)

Descriptive Statistics:

Descriptive Statistics:						
	count	mean	std	min	25%	50% \
OPEN	473.0	2.588288e+03	3.857626e+02	1229.40	2388.2	2552.50
close	473.0	2.586382e+03	3.865337e+02	1223.00	2383.7	2552.55
HIGH	473.0	2.608841e+03	3.903360e+02	1243.95	2400.7	2573.50
LOW	473.0	2.565801e+03	3.825046e+02	1217.25	2372.2	2532.00
VOLUME	473.0	6.792192e+06	3.829456e+06	213020.00	4337938.0	5712541.00
	75%	max				
OPEN	2918.0	3208.9				
close	2917.3	3201.8				
HIGH	2944.1	3217.6				
LOW	2894.7	3171.0				
VOLUME	8211221.0	28923341.0				

Figure: 2 Descriptive Statistics

The descriptive statistics provide a summary of Reliance Equity's stock price and trading volume of over 473 trading days. The CLOSE price has a mean of 2586.83 INR and a median of 2552.55 INR, indicating that the stock price distribution is relatively centered but shows notable variability. The standard deviation of approximately 386 INR highlights significant price fluctuations, with prices ranging from a minimum of 1223 INR to a maximum of 3201.8 INR. The OPEN, HIGH, and LOW prices follow similar trends, with maximum values for HIGH reaching 3217.6 INR and LOW bottoming at 1217.25 INR. These observations suggest periods of market volatility and investor activity. The VOLUME of shares traded averages at 6.79 million, but with a substantial standard deviation of 3.8 million, it indicates trading activity spikes, peaking at 28.9 million shares. Such high trading volumes often align with major market events or shifts in investor sentiment.

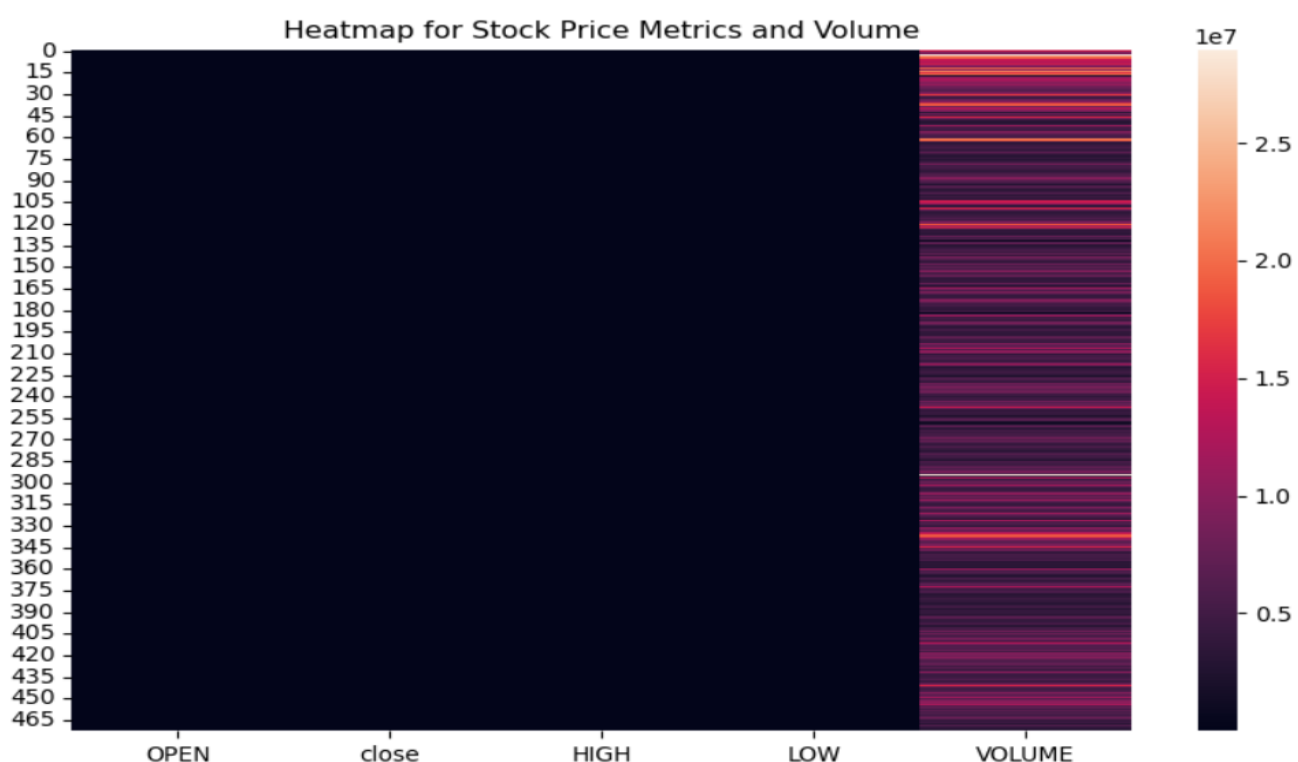


Figure:3 Heatmap for descriptive statistics

The heatmap displays correlations between price metrics (OPEN, CLOSE, HIGH, LOW) and trading VOLUME. While VOLUME shows variability, its strong alignment with price metrics shows that trading activity influences price movements, supporting further analysis of volume-price relationships and these insights form the foundation for analysing price trends, volatility, and trading behaviour.

1. What are the trends in the daily stock price (OPEN, CLOSE, HIGH, and LOW) of Reliance Equity over the given period?

The trends in Reliance Equity's stock price over the given period were analysed using line plots for the OPEN, CLOSE, HIGH, and LOW prices. The plot shows that stock prices experienced fluctuations with noticeable upward and downward trends.

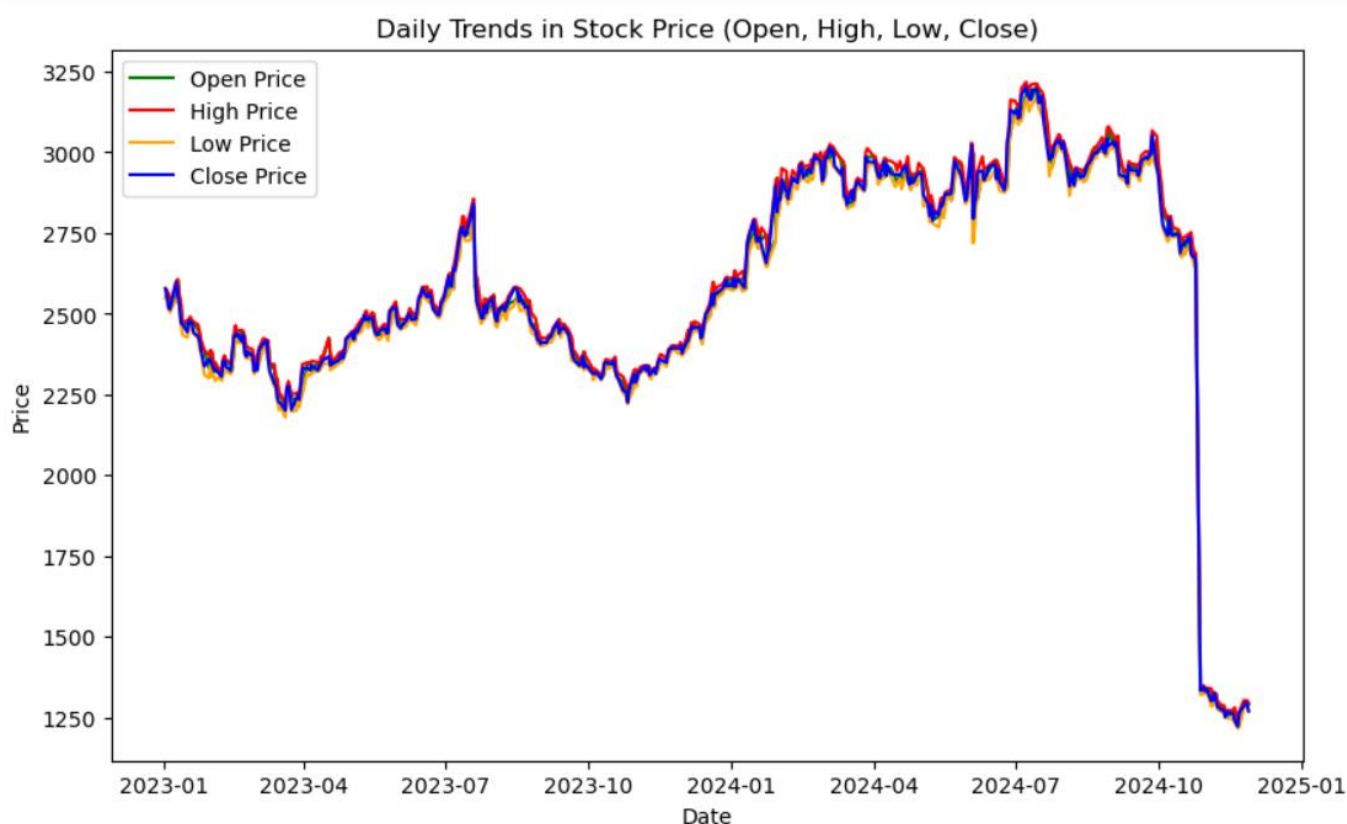


Figure:4 Daily stock trends over time

The stock price remained relatively stable during early 2023, followed by a significant uptrend starting in mid-2023 and peaking near October 2023. However, a sharp decline in prices is observed towards the end of 2024, indicating a possible market correction or external influencing factor. Throughout the period, the HIGH and LOW prices closely follow the CLOSE price, reflecting consistent intraday volatility.

In addition to stock price trends, trading VOLUME was analysed over the same period. The trading volume shows frequent spikes, with some reaching over 25 million shares. These spikes align with significant price movements which suggests increased investor activity during periods of market volatility or news events. Higher trading volumes often coincide with sharp price movements which indicates investor interest and liquidity. Periods of low volume reflects reduced market participation which contributes to slower price changes.

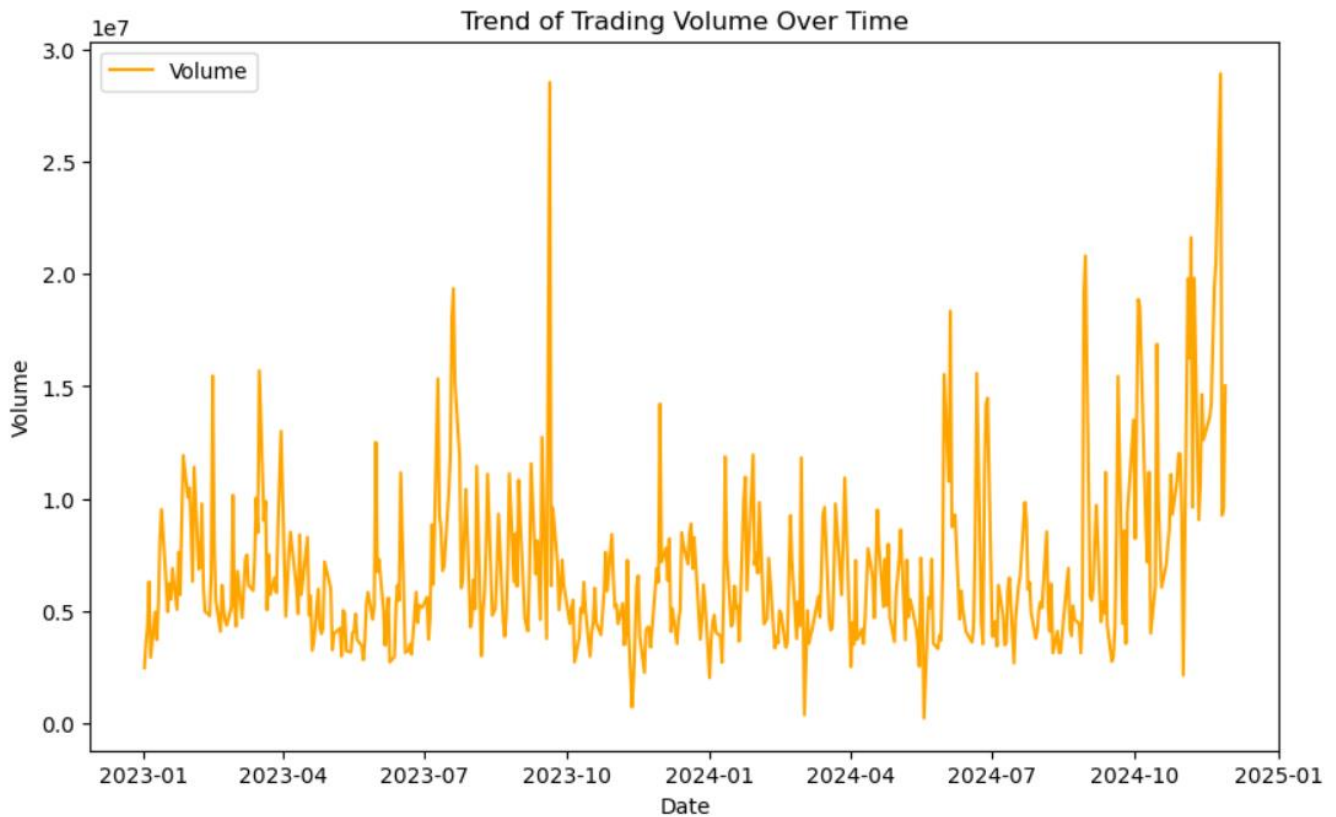


Figure: 5 Line Plot for volume over time

How does trading volume influence stock price movements?

The relationship between trading volume and key stock price metrics (OPEN, HIGH, LOW, CLOSE) was analysed using scatterplots. This analysis explores how trading activity influences price movements.

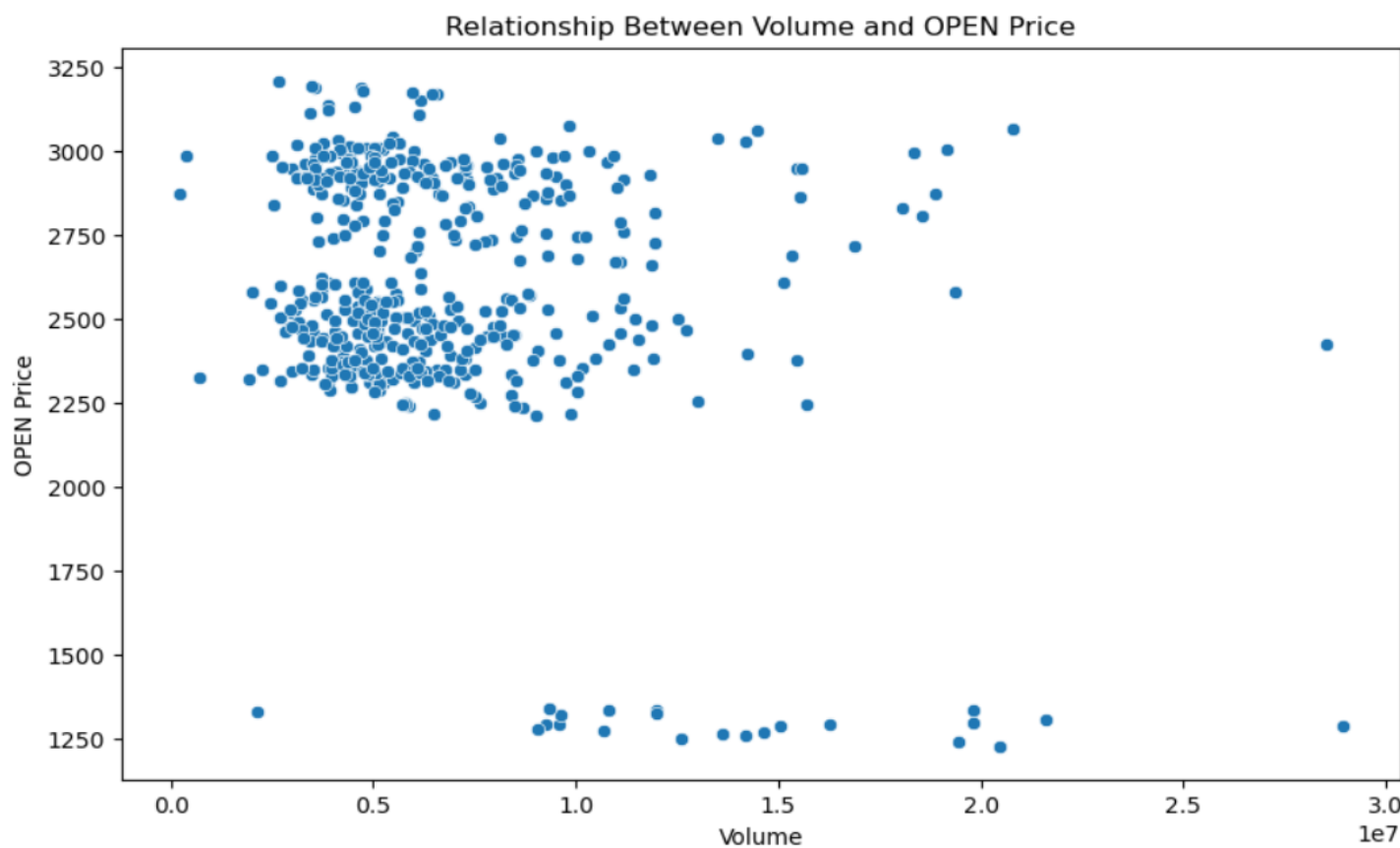


Figure: 6 Scatter plot showing relationship between volume and open price.

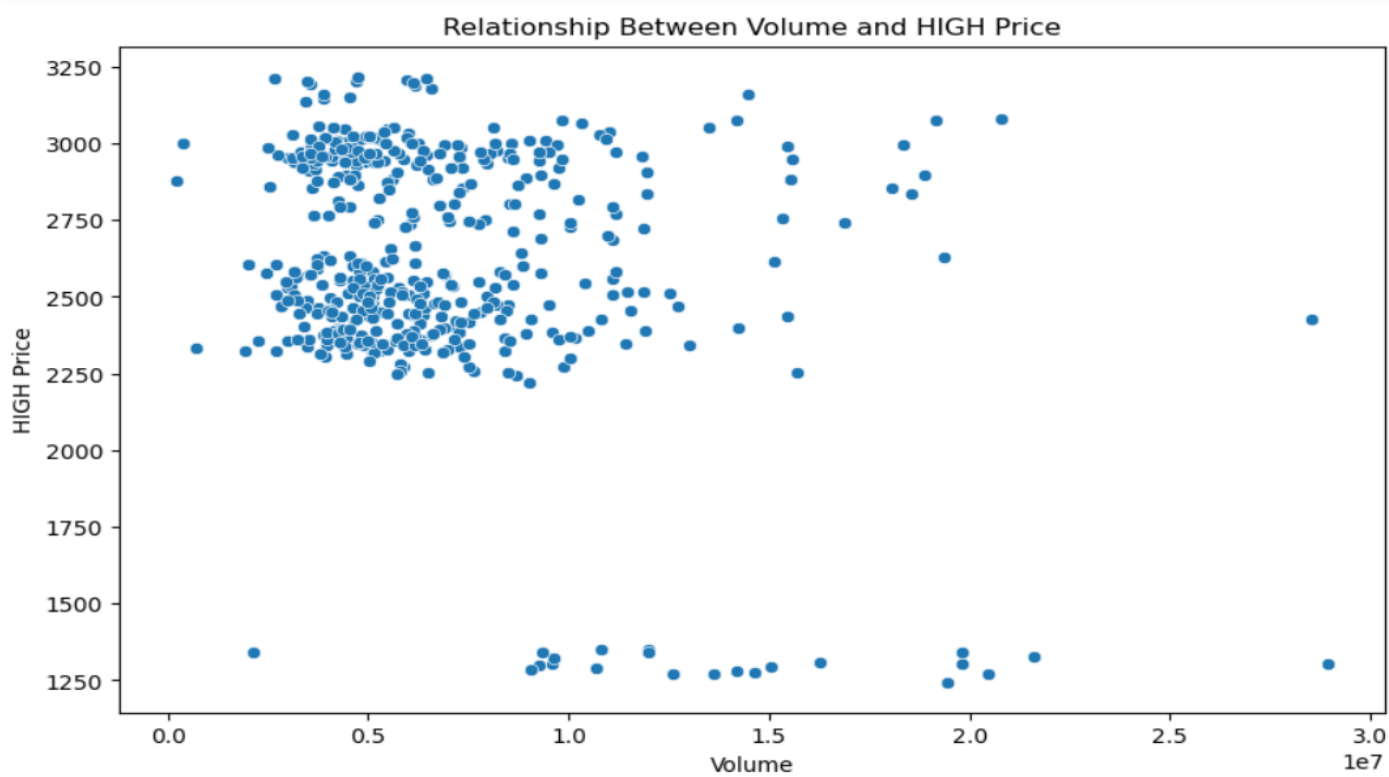


Figure: 7 Scatter plot showing relationship between volume and high price.

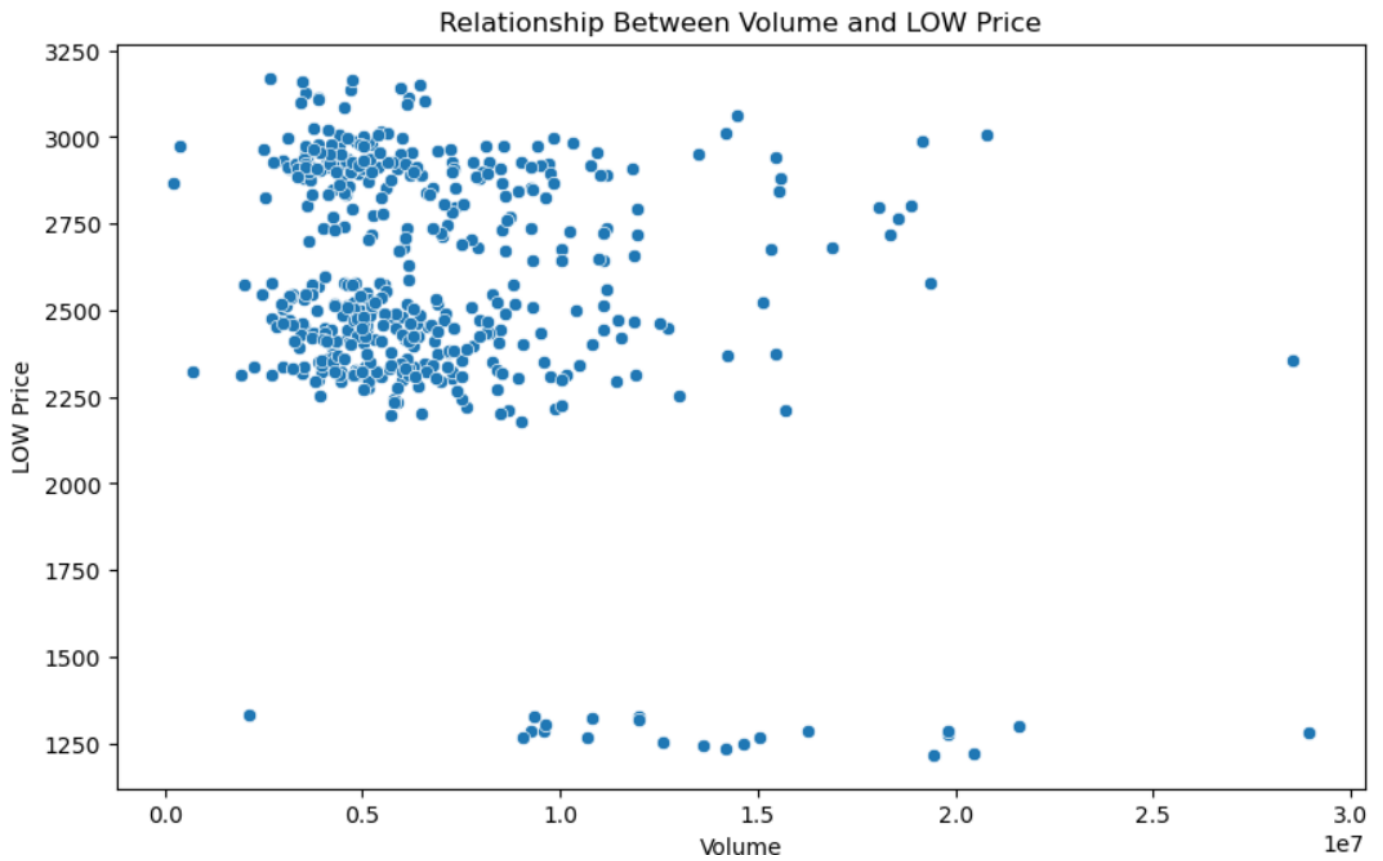


Figure: 8 Scatter plot showing relationship between volume and low price.

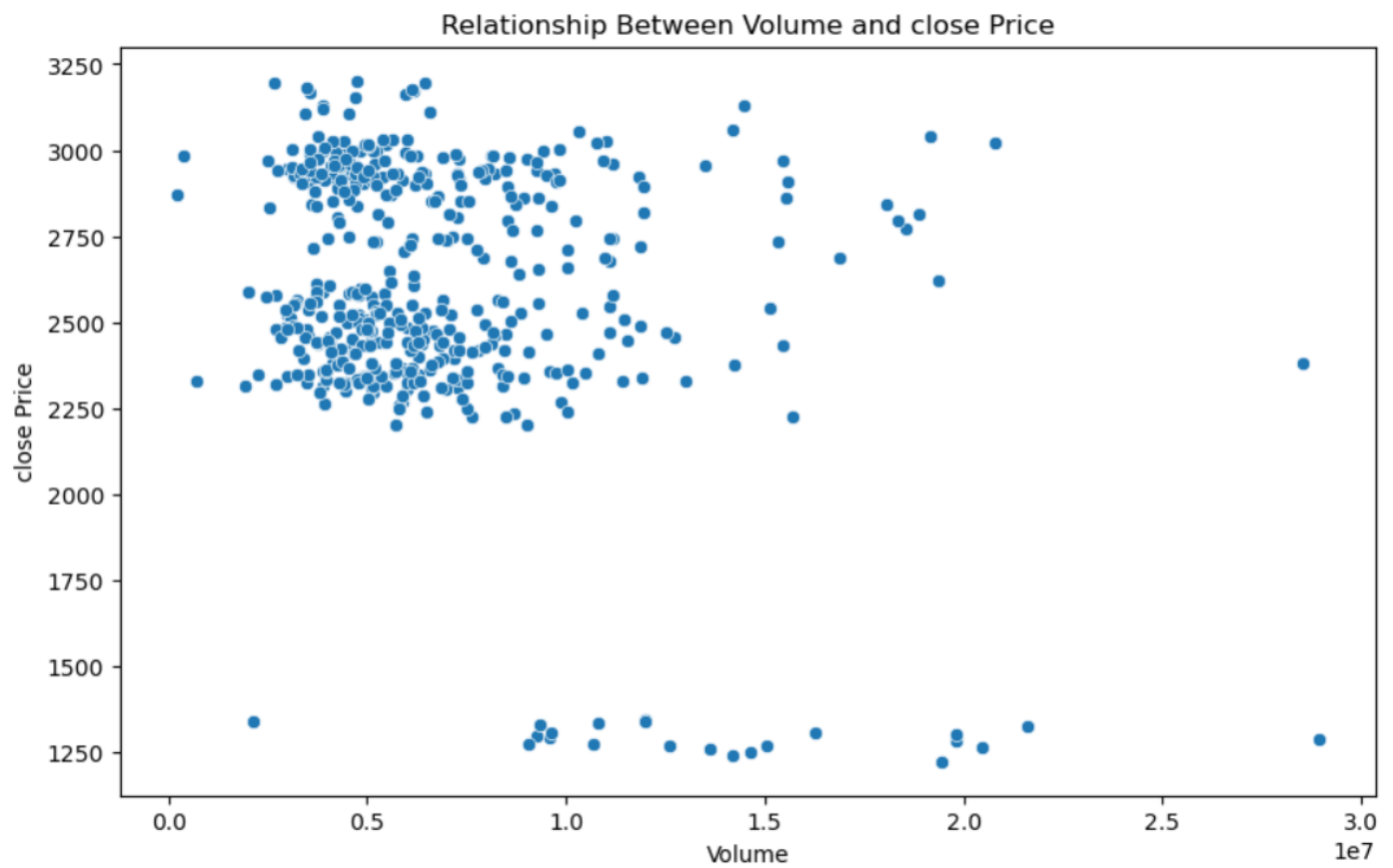


Figure: 9 Scatter plot showing relationship between volume and close price.

The scatterplots display clusters around two price ranges: 2500–2750 INR and 3000–3200 INR, suggesting that the stock experienced higher trading activity and liquidity at these levels. When trading VOLUME exceeded 10 million shares, there seen a greater variability in stock prices, reflecting the influence of significant investor activity, market events, or news. While the relationship between trading volume and price movements shows a positive trend, it is not strictly linear. Higher trading activity tends to align with notable price changes, indicating market volatility during these periods. Moreover, Surges in trading volume often accompany significant price fluctuations, which can signal market sentiment shifts or reactions to external triggers. Instances where high volumes occur at lower price levels may indicate sell-offs or market corrections.

What significant peaks or troughs can be identified, and what insights do they provide for trading strategies?

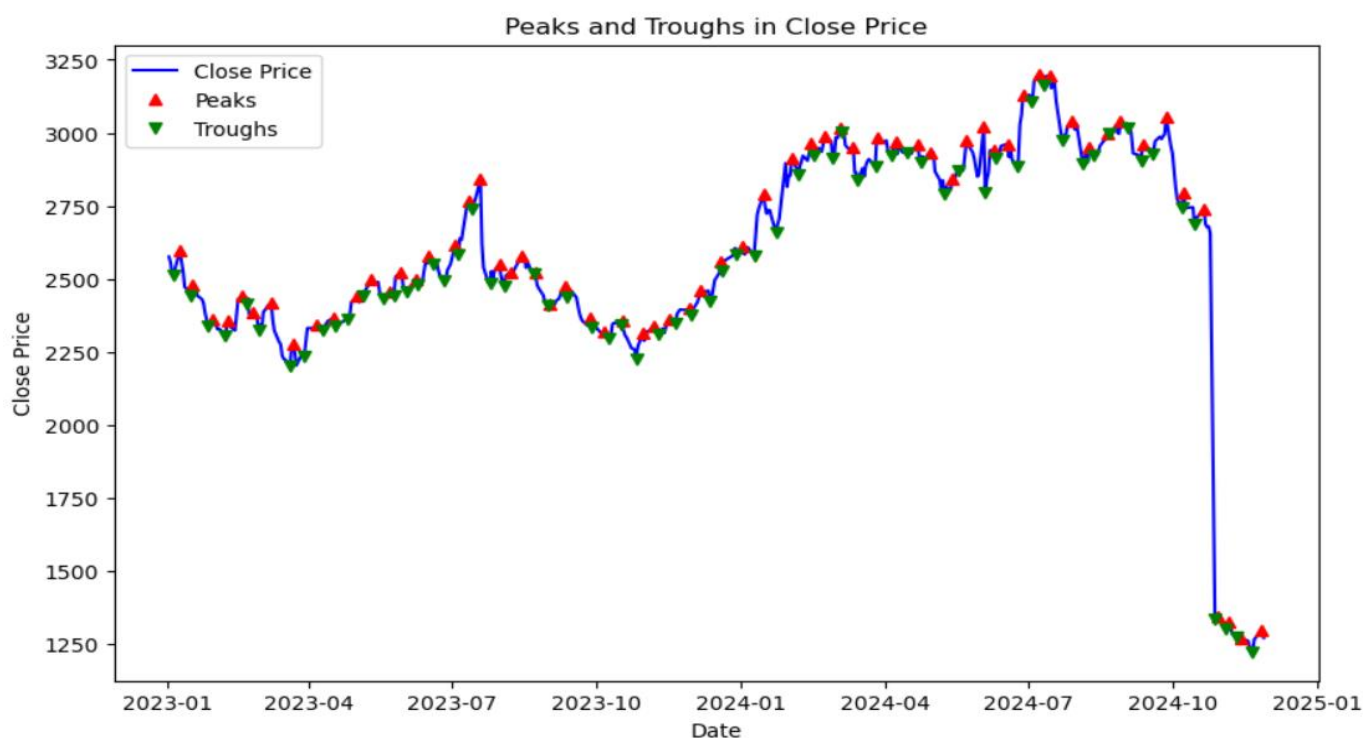


Figure: 10 Peaks and troughs in close price

The analysis of Close Price reveals significant peaks (highlighted as red triangles) and troughs (green triangles), marking key turning points in the stock's price movement. Peaks represent moments where the stock price reached a temporary high point before declining, while troughs indicate points where the price hit a low point before bouncing back. The line chart shows multiple peaks and troughs, particularly during mid-2023 and early 2024, where price movements were more volatile. A noticeable peak around mid-2023 aligns with a sharp upward trend which suggests strong buying interest. On the contrary, troughs during late 2023 and late 2024 highlight periods of price corrections and market pullbacks. Peaks can act as potential resistance levels, where traders may consider selling to maximise profits whereas troughs serve as support levels, presenting opportunities for investors to buy at lower prices in anticipation of recovery. By understanding these critical turning points, traders can optimise their entry and exit strategies, helping them navigate price trends and maximise returns.

What are the statistical relationships between price metrics and trading volume?

The relationship between trading volume and key price metrics (OPEN, CLOSE, HIGH, and LOW) is examined using a correlation heatmap and pair plots to uncover patterns and associations.

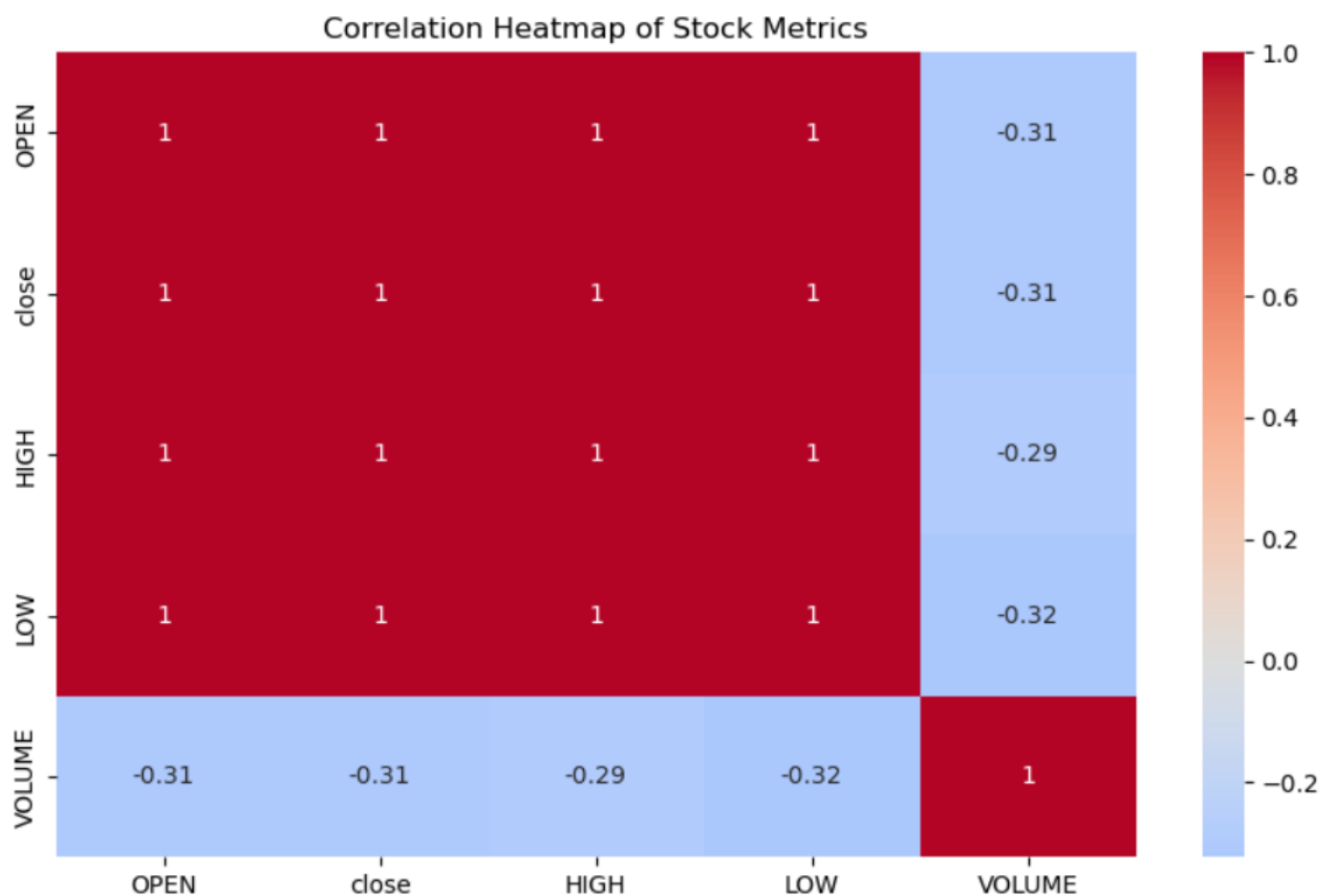


Figure: 11 Correlation Heatmap of stock metrics

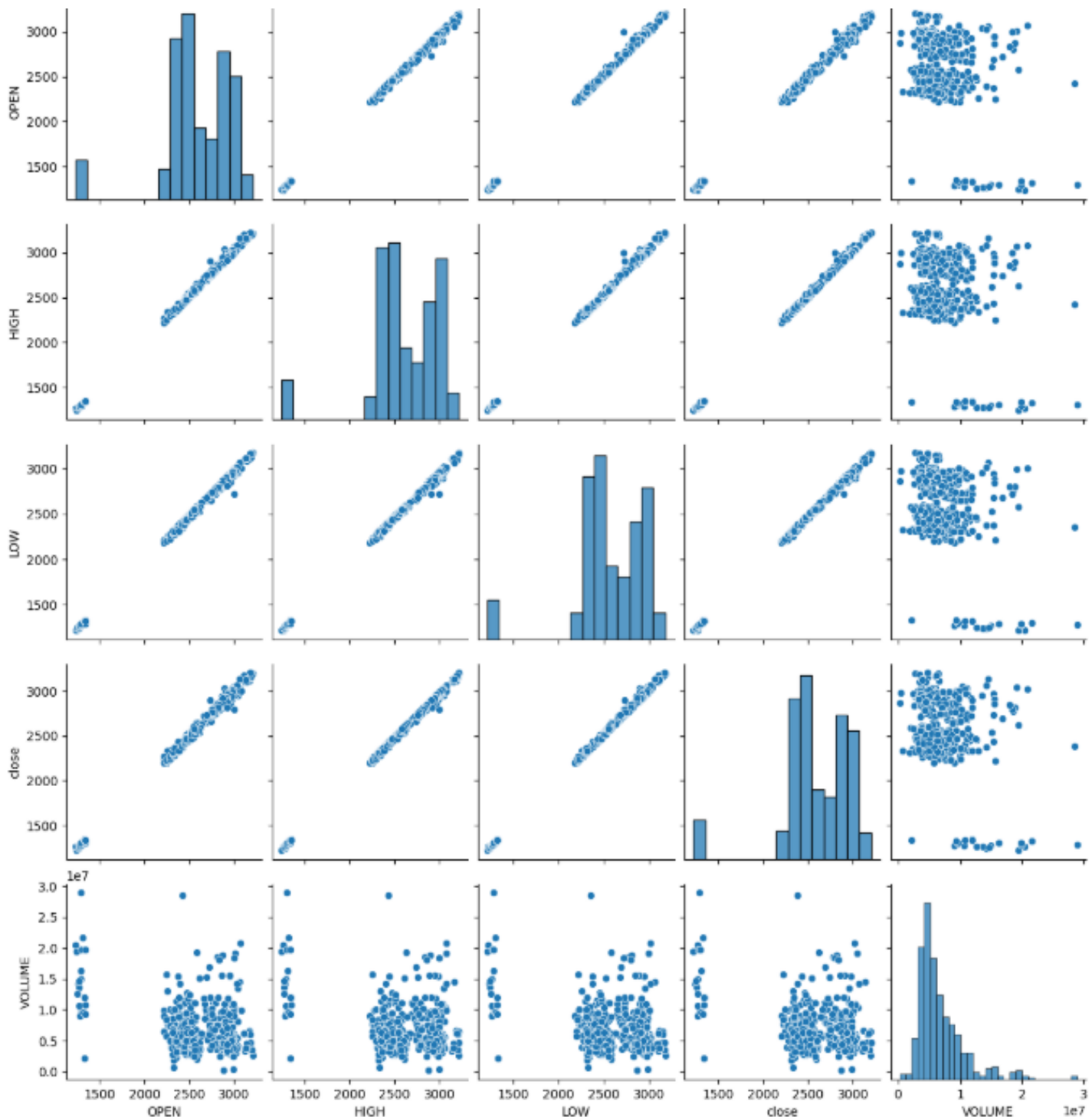


Figure: 12 Pair plotting of stock metrics

The heatmap displays a strong positive correlation between all price metrics which indicates that OPEN, CLOSE, HIGH, and LOW prices move closely together. However, trading VOLUME shows a weak negative correlation with price metrics, ranging between -0.29 and -0.32 . This suggests that higher trading volumes often coincide with price declines, possibly reflecting sell-offs or market corrections. The pair plots provide a deeper visual understanding of the price metrics which shows clear linear relationships, confirming their strong interdependence. VOLUME exhibits scattered patterns with price metrics, reinforcing the weak correlation and highlighting that spikes in trading activity are not always associated with price increases. These findings indicate that while price metrics move predictably together, trading volume behaves differently, and often reflecting market volatility and investor sentiment rather than direct price influence.

What is the volatility in stock prices, and how can it inform investor risk management strategies?

Stock price volatility is analysed using Bollinger Bands, which consists of the upper band, lower band, and the Close Price. These bands provide a visual representation of price fluctuations and potential market trends over time.

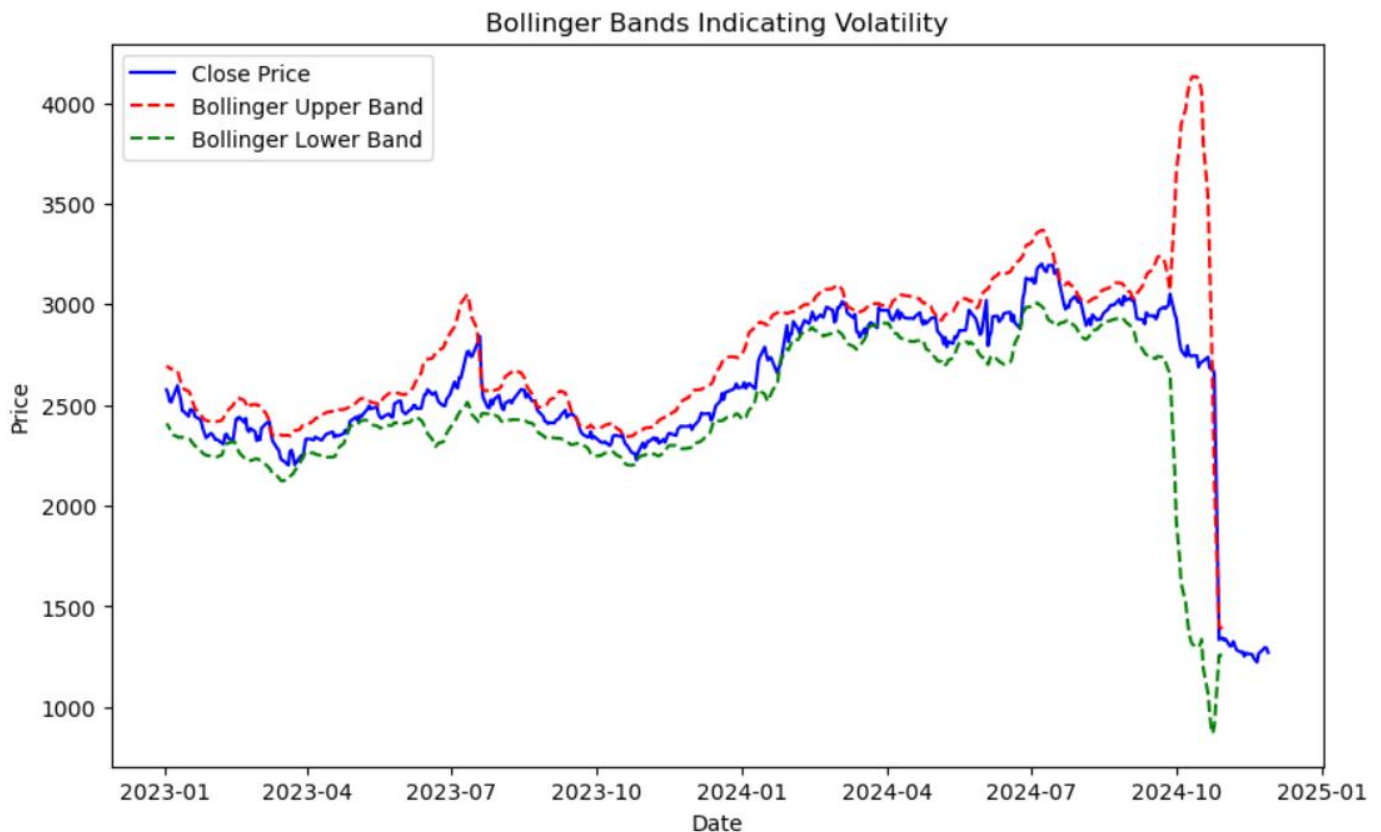


Figure: 13 Pair plotting of stock metrics

When the Close Price (blue line) comes near the upper Bollinger Band (red dashed line), it indicates that the stock may be overbought, signalling a potential downward correction. On the contrary, when the Close Price approaches the lower band (green dashed line), it suggests the stock may be oversold, indicating a potential buying opportunity. Notably, during late 2024, there is an extreme widening of the Bollinger Bands which reflects heightened volatility and significant market uncertainty. Investors can use Bollinger Bands to identify periods of high volatility and adjust their strategies, such as exiting positions during extreme price movements or entering trades when prices stabilise near the lower band. These bands also help in managing risk by signalling market entry and exit points, allowing investors to act before significant reversals occur.

Analytics Techniques Used

Correlation Analysis:

Correlation Analysis:

	OPEN	close	HIGH	LOW	VOLUME
OPEN	1.000000	0.996802	0.998694	0.998275	-0.306304
close	0.996802	1.000000	0.998795	0.998788	-0.308234
HIGH	0.998694	0.998795	1.000000	0.998288	-0.294291
LOW	0.998275	0.998788	0.998288	1.000000	-0.322705
VOLUME	-0.306304	-0.308234	-0.294291	-0.322705	1.000000

Figure: 14 Correlation Analysis

The correlation analysis explores relationships between key price metrics (OPEN, CLOSE, HIGH, LOW) and VOLUME. Price metrics (OPEN, CLOSE, HIGH, LOW) shows a strong positive correlation, with values exceeding 0.99, indicating they move closely together and this suggests that changes in one price metric strongly influence the others. Trading VOLUME shows a weak negative correlation with price metrics, ranging from -0.29 to -0.32 which indicates that higher trading volumes often accompany declining prices, possibly reflecting sell-offs or increased volatility. The strong interdependence among price metrics highlights consistent market behaviour, while the negative correlation with volume suggests that trading surges may not always drive price increases. Understanding these relationships helps investors gauge market sentiment and forecast potential price movements.

How does the stock price's moving average behave over time, and what trends can be observed?

The stock price's behaviour over time was analysed using 7-day, 30-day, 50-day, and 100-day moving averages, which provide insights into short, medium, and long-term trends.

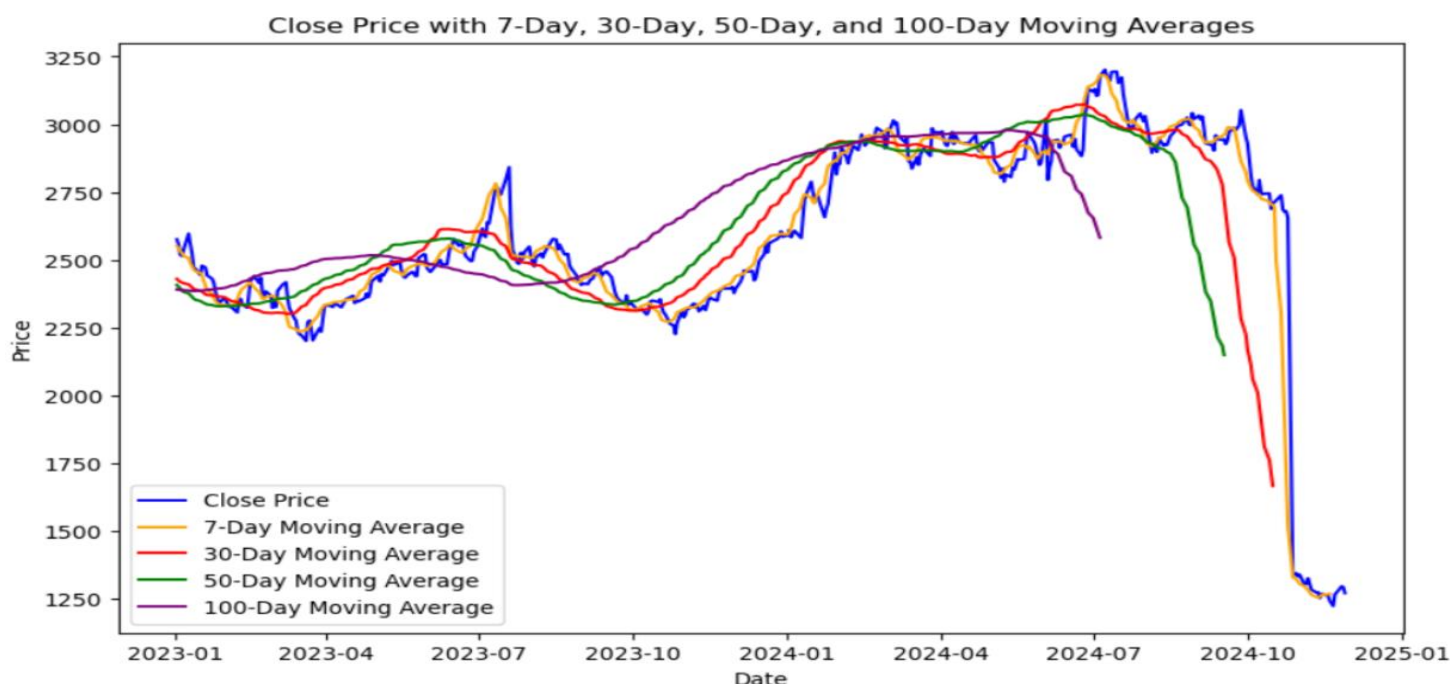


Figure: 15 Time-Series Analysis

The 7-day moving average (orange line) closely tracks the Close Price (blue line), showing how quickly it reacts to daily price movements. Sharp upward spikes in mid-2023 and early 2024, as well as steep declines in late 2024, are captured effectively. This moving average is ideal for detecting short-term momentum changes.

The 30-day (red line) and 50-day (green line) moving averages provide a clear view of intermediate trends. They smooth out smaller fluctuations and confirm significant price movements, such as the recovery from mid-2023 lows and the peak seen in mid-2024.

The 100-day moving average (purple line) shows a stable, long-term perspective. It remained upward-sloping for most of the analysed period which reflects a sustained growth, however it started declining sharply towards the end of 2024, alerting a market correction.

Crossovers between shorter and longer moving averages often indicate trend reversals. For instance, downward crossovers in late 2024 align with the steep drop in stock prices. Moving averages help investors study market direction, filter out noise, and make informed decisions about buying or selling stocks. Overall, this analysis highlights the importance of moving averages in identifying trends, predicting reversals, and managing trading strategies.

Predictive Analysis:

```
Mean Squared Error: 32120.308138523196
R-squared Score: 0.8121616956743778
Regression Coefficients:
  Feature  Coefficient
0  VOLUME -1.766562e-04
1   VALUE  7.049184e-08
```

Figure: 16 Predictive Analysis

A simple regression model is developed to predict the Close Price based on VOLUME and VALUE. The model's performance is evaluated using the Mean Squared Error (MSE) and the R-squared score, with results Mean Squared Error: 32,120, indicating the average squared difference between the predicted and actual Close Prices. A lower MSE suggests a better model fit. R-squared Score: 0.812, means the model explains approximately 81.2% of the variability in the Close Price. This suggests that the selected features, VOLUME and VALUE, contribute significantly in predicting stock prices. VOLUME: -1.7665e-04, this negative coefficient indicates that as trading volume increases, the Close Price tends to decrease slightly, possibly due to market sell-offs. VALUE: 7.0491e-08, this positive coefficient suggests that higher trade values are associated with an increase in the Close Price. The predictive analysis highlights a moderate relationship between trading activity and the stock price.

Reflective Evaluation of the report:

This project offered a valuable opportunity to apply business analytics techniques to a real-world dataset which allowed me to enhance both my technical abilities and analytical reasoning. By exploring Reliance Equity's stock data, I was able to analyse trends, volatility, and the relationship between trading volume and price metrics, answering critical business questions and providing actionable insights. Through descriptive statistics and visualisations, I gained a clear understanding of how stock prices behaved over time. Key findings, such as identifying peaks and troughs in the Close Price, revealed significant turning points in market performance. The Bollinger Bands effectively highlighted periods of heightened volatility, particularly during late 2024, while the moving averages offered clarity on short, medium, and long-term trends. The predictive analysis further showcased the influence of VOLUME and VALUE on the Close Price, achieving a strong R-squared score of 81.2%, demonstrating the value of regression models for forecasting.

This project has taught me how to handle data systematically, from preparation and cleaning to applying advanced analytical techniques and interpreting the results. It focuses on the importance of accuracy, ethical considerations, and ensuring my findings aligned with business objectives. Overall, this project highlighted the critical role of a business analyst in turning complex datasets into meaningful insights. By connecting analysis to real-world scenarios, I have developed skills in storytelling, visualisation, and decision-making. This experience has boosted my confidence in dealing analytical challenges and presenting insights that support informed business strategies.

Conclusion

This report successfully analysed Reliance Equity's stock performance using a variety of business analytics techniques, providing a clear understanding of stock price trends, volatility, and the influence of trading volume. Through systematic data preparation, cleaning, and exploration, valuable insights are derived. The descriptive statistics and visualisations highlighted key patterns, such as peaks, troughs, and significant periods of price movement, which are crucial for trading strategies. Bollinger Bands shows periods of volatility, offering insights for risk management, while moving averages effectively captured short and long-term trends in stock price behaviour. Additionally, the predictive analysis displays the ability to forecast stock prices using VOLUME and VALUE, with an R-squared score of 81.2%, showing the strength of analytical models in real-world applications.

Overall project, shows the importance of data-driven approaches in addressing business challenges. By applying statistical analysis, visualisation tools, and predictive models, the report provided actionable findings that can help stakeholders in making informed decisions and navigate market uncertainties. This project highlights the role of a business analyst in transforming data into meaningful insights. It emphasised technical proficiency, critical thinking, and the ability to communicate findings effectively. The skills and knowledge gained through this project are invaluable for future analytical work, particularly in driving business strategy, identifying opportunities, and managing risks in dynamic markets.

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Appendix

[A] Data Cleaning and Preparation

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

```
•[3]: # Step 1: Load and Prepare the Data
file_path = r"C:\Users\91942\Downloads\Equity Reliance.xlsx"
df = pd.read_excel(file_path, sheet_name='Quote-Equity-RELIANCE-EQ-01-01-')
df.head(26)
```

```
•[6]: # Step 2: Data Cleaning
df.columns = df.columns.str.strip()
```

```
•[7]: # Converting 'Date' column to datetime format
df['Date'] = pd.to_datetime(df['Date'])
```

```
•[8]: # Checking for missing values and drop any rows with missing data
df = df.dropna()
```

```
•[9]: # Removing duplicates if any
df = df.drop_duplicates()
```

```
[12]: # Creating new derived metrics
df['Price Range'] = df['HIGH'] - df['LOW']
df['Daily Change'] = df['close'] - df['OPEN']
```

[B] Descriptive Statistics and Visualisations

```
[14]: # Step 3: Descriptive Statistics
# Calculate descriptive statistics for relevant columns
descriptive_stats = df[['OPEN', 'close', 'HIGH', 'LOW', 'VOLUME']].describe().T
```

```
[47]: # Displaying descriptive statistics
print("Descriptive Statistics:\n", descriptive_stats)
```

```
[48]: # Visualising descriptive statistics
plt.figure(figsize=(10, 6))
sns.heatmap(data=df[['OPEN', 'close', 'HIGH', 'LOW', 'VOLUME']])
plt.title('Heatmap for Stock Price Metrics and Volume')
plt.show()
```

[C] Daily stock trends over time

```
# Step 4: Stock Trends Over Time Q1
plt.figure(figsize=(10, 6))
plt.plot(df['Date'], df['OPEN'], label='Open Price', color='green')
plt.plot(df['Date'], df['HIGH'], label='High Price', color='red')
plt.plot(df['Date'], df['LOW'], label='Low Price', color='orange')
plt.plot(df['Date'], df['close'], label='Close Price', color='blue')
plt.xlabel('Date')
plt.ylabel('Price')
plt.title('Daily Trends in Stock Price (Open, High, Low, Close)')
plt.legend()
plt.show()
```

[D] Line plot for volume over time

```
[24]: # Line plot for VOLUME over time
plt.figure(figsize=(10, 6))
plt.plot(df['Date'], df['VOLUME'], label='Volume', color='orange')
plt.xlabel('Date')
plt.ylabel('Volume')
plt.title('Trend of Trading Volume Over Time')
plt.legend()
plt.show()
```

[E] Scatter plots for volume vs all key price metrics

```
# Scatterplots for VOLUME vs all key price metrics Q2
for col in ['OPEN', 'HIGH', 'LOW', 'close']:
    plt.figure(figsize=(10, 6))
    sns.scatterplot(x=df['VOLUME'], y=df[col])
    plt.xlabel('Volume')
    plt.ylabel(f'{col} Price')
    plt.title(f'Relationship Between Volume and {col} Price')
    plt.show()
```

[F] Peaks and troughs in close price

```
[44]: # Analysis of 52W Highs and Lows Q3
from scipy.signal import find_peaks

# Detect peaks and troughs in CLOSE price
peaks, _ = find_peaks(df['close'], distance=5)
troughs, _ = find_peaks(-df['close'], distance=5)

plt.figure(figsize=(10, 6))
plt.plot(df['Date'], df['close'], label='Close Price', color='blue')
plt.plot(df['Date'].iloc[peaks], df['close'].iloc[peaks], 'r^', label='Peaks')
plt.plot(df['Date'].iloc[troughs], df['close'].iloc[troughs], 'gv', label='Troughs')
plt.xlabel('Date')
plt.ylabel('Close Price')
plt.title('Peaks and Troughs in Close Price')
plt.legend()
plt.show()
```

[G] Correlation Heatmap and pair plot

```
[43]: # Step 5: Volume-Price Relationship Q4
# Correlation heatmap
sns.pairplot(df[['OPEN', 'HIGH', 'LOW', 'close', 'VOLUME']])
plt.figure(figsize=(10, 6))
correlation_matrix = df[['OPEN', 'close', 'HIGH', 'LOW', 'VOLUME']].corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', center=0)
plt.title('Correlation Heatmap of Stock Metrics')
plt.show()
```

[H] Rolling standard deviation and Bollinger Bands

```
[49]: # Calculating rolling standard deviation and Bollinger Bands Q5
df['Rolling Std'] = df['close'].rolling(window=20).std()
df['Bollinger Upper'] = df['7-day MA'] + (df['Rolling Std'] * 2)
df['Bollinger Lower'] = df['7-day MA'] - (df['Rolling Std'] * 2)

plt.figure(figsize=(10, 6))
plt.plot(df['Date'], df['close'], label='Close Price', color='blue')
plt.plot(df['Date'], df['Bollinger Upper'], label='Bollinger Upper Band', color='red', linestyle='--')
plt.plot(df['Date'], df['Bollinger Lower'], label='Bollinger Lower Band', color='green', linestyle='--')
plt.xlabel('Date')
plt.ylabel('Price')
plt.title('Bollinger Bands Indicating Volatility')
plt.legend()
plt.show()
```

[I] Correlation Analysis

```
[31]: # Step 7: Correlation Analysis
# Correlation between price metrics and volume
correlation_results = df[['OPEN', 'close', 'HIGH', 'LOW', 'VOLUME']].corr()
print("Correlation Analysis:\n", correlation_results)
```

[J] Time Series Analysis

```
[41]: # Step 6: Analytics Techniques and Models Q6
# Time Series Analysis (Moving Averages)
df['7-day MA'] = df['close'].rolling(window=7).mean()
df['30-day MA'] = df['close'].rolling(window=30).mean()
df['50-day MA'] = df['close'].rolling(window=50).mean()
df['100-day MA'] = df['close'].rolling(window=100).mean()

# Plot the close price with moving averages
plt.figure(figsize=(10, 6))
plt.plot(df['Date'], df['close'], label='Close Price', color='blue')
plt.plot(df['Date'], df['7-day MA'], label='7-Day Moving Average', color='orange')
plt.plot(df['Date'], df['30-day MA'], label='30-Day Moving Average', color='red')
plt.plot(df['Date'], df['50-day MA'], label='50-Day Moving Average', color='green')
plt.plot(df['Date'], df['100-day MA'], label='100-Day Moving Average', color='purple')
plt.xlabel('Date')
plt.ylabel('Price')
plt.title('Close Price with 7-Day, 30-Day, 50-Day, and 100-Day Moving Averages')
plt.legend()
plt.show()
```

[K] Predictive Analysis

```
[32]: # Step 8: Predictive Analytics (Simple Regression Model)
      from sklearn.model_selection import train_test_split
      from sklearn.linear_model import LinearRegression
      from sklearn.metrics import mean_squared_error, r2_score

      # Define feature (X) and target (y) variables
      X = df[['VOLUME', 'VALUE']]
      y = df['close']

      # Split data into training and testing sets (80% train, 20% test)
      X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

      # Create and fit the regression model
      reg_model = LinearRegression()
      reg_model.fit(X_train, y_train)

      # Predict on the test data
      y_pred = reg_model.predict(X_test)

      # Model evaluation
      mse = mean_squared_error(y_test, y_pred)
      r2 = r2_score(y_test, y_pred)
      print("Mean Squared Error:", mse)
      print("R-squared Score:", r2)

      # Display regression coefficients
      print("Regression Coefficients:\n", pd.DataFrame({'Feature': X.columns, 'Coefficient': reg_model.coef_}))
```

Thank You.